** Surveying **

* Introduction:-

→ It is the art of determining relative positions of points through direct or indirect measurement of distance, direction and elevation. It is also Include establishment of point on ground which pre-determined in paper.

* Types of surveying:-

- on the basis of whether the curvature of the earth is taken into account or not, surveying can be divided into two main categories.

i). Plane surveying:-

→ plane surveying is that in which effect of earth curvature is not consider.

→ It is sutaible for long area [< 250 Km²].

ii). Geodatic survey:-

→ Geodatic survey is that in which effect of earth curvature is consider.

→ It is sutaible for large area [> 250 Km²].

- Classification on the basis of surface and area surveyed:-

1) Land survey:-

→ Land surveys are done for objects on the surface of the earth. It can be subdivided into:

a) Topographic Survey:-

→ This is for depicting the (hills, valleys, mountains, rivers) etc. and manmade features (roads, houses, settlements) on the surface of the earth.

b) Cadastral Survey:-

→ Cadastral survey is used to determining property boundaries including those of fields, houses, plots of land, etc.

c) Engineering Survey:-

→ Engineering survey is used to acquired data for the planning, design and Execution of Engineering project like roads, bridges, Canals, dams, railways, buildings etc.

d) City Survey:-

→ The surveys involving the Construction and developement of towns including roads, drainage, water supply, sewage street network, etc. are generally referred to as city survey.

2) Marine or Hydrographic Survey:-

→ Those are type of survey in which surveys of large water bodies for navigation, tidal monitoring the Construction of harbours etc.

3) Astronomical Survey:-

→ Astronomical survey uses the observations of the heavenly bodies (sun, moon, stars etc). to fix the

absolute locations of places on the surface of earth.

* Principles of Surveying:-

→ To locate the position of a point by measurement from two reference point.

→ To work from whole to part.

→ Two control point are selected in an area and the distance between them is measured accurately.

→ In working from whole to part major control points are selected and measured first with high degree of precision, minor details can be collected after or later on even with less degree of precision. In this manner, errors involved in minor detailing will be compensating and will not affect the major dimensions.

Note:- i) The main idea of working from whole to part is to localise errors and prevent their accumulation.

ii). on the contrary, working from part to whole the error accumulate and expand to greater magnitude and the survey work becomes uncontrollable at the end.

* Site plans:-

→ A site plan - sometimes called a plot plan - is an architectural document that functions as a readable map of a building site, giving you all the details you need to know about how the structure will be

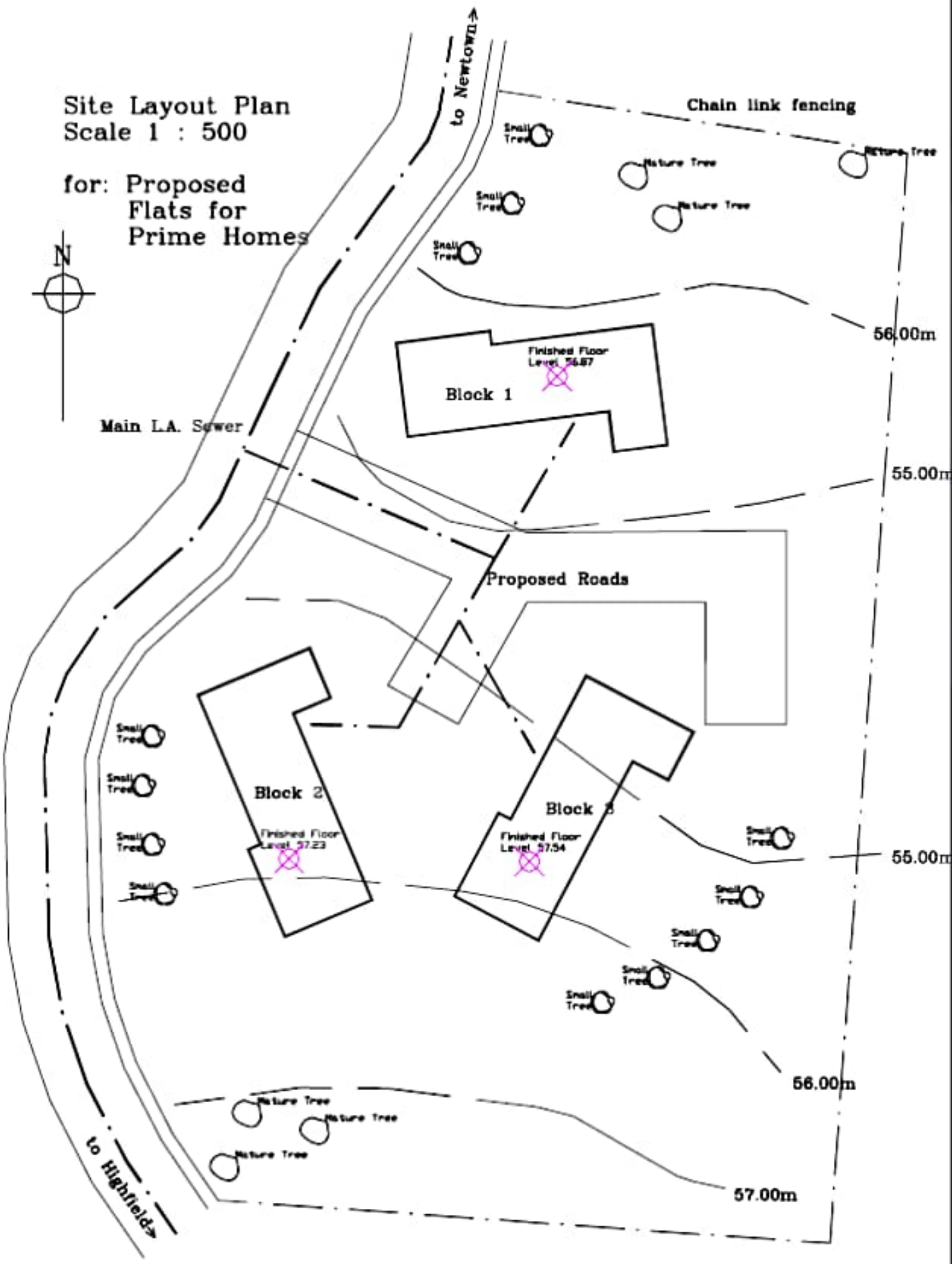
oriented on the lot.

→ A builder or contractor will create a diagram that shows the plot of land and its property lines, along with its landscape features, structural elements, setbacks, utility poles and power lines, fencing and on-site structures.

→ Real estate agent use site plans to show all the home's key features, including the size, scale, and orientation and key outdoor features.

→ Information should be included in site plan:-

i) property line
ii) Distance between building and property line
iii) Existing and proposed condition
iv) outline of existing and proposed building and structures.
v) Distance between buildings.
vi) parking lot, indicating parking space
vii) Surrounding streets.
viii) ground sign location
ix) Utilities.


Site Layout Plan
Scale 1 : 500
for: Proposed
Flats for
Prime Homes
N
to Newtown
Chain link fencing
Small Tree
Mature Tree
Finished Floor Level 56.87
Block 1
56.00m
Main L.A. Sewer
55.00m
Proposed Roads
Block 2
Finished Floor Level 57.23
Block 3
Finished Floor Level 57.54
55.00m
56.00m
57.00m
to Highfield